TIC 295254637

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       2.664955 d  (ALIASED, see below)
    -> corrected period: 5.329910 d
  Epoch (t0)   1626.9366 BTJD
  Duration     0.1200 d
  Depth        3.867 %  (38,674 ppm)
  SNR          176.9

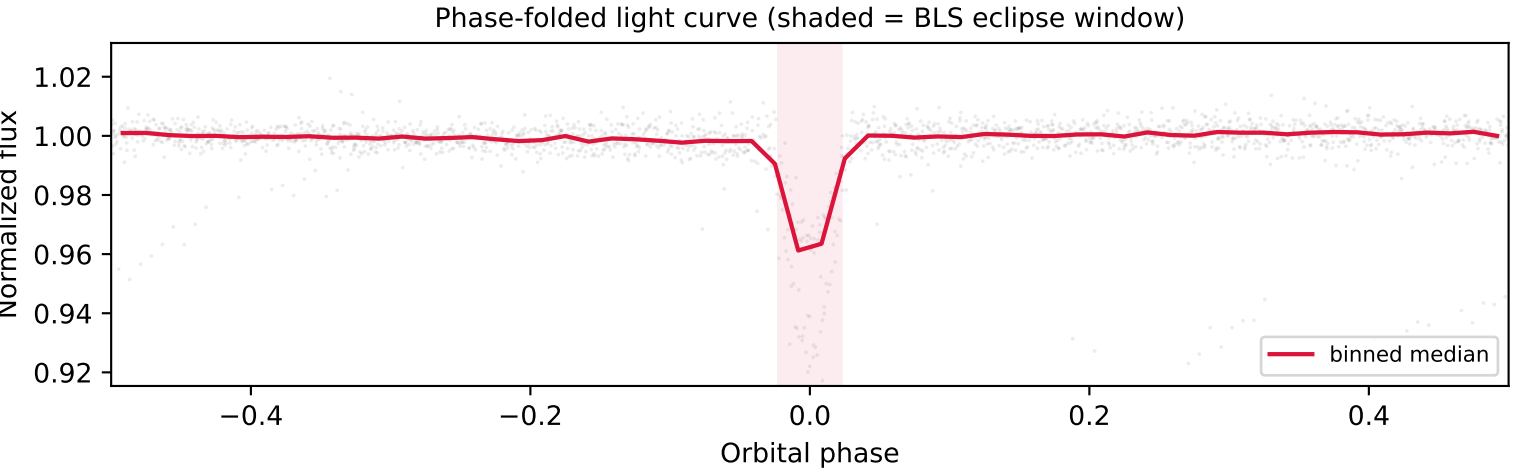
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     17  (gate >= 3)
  Robust depth ratio    0.87  (gate >= 0.50)
  Robust depth sigma    74.0  (gate >= 7.0)
  Odd/even z-score      15.37  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

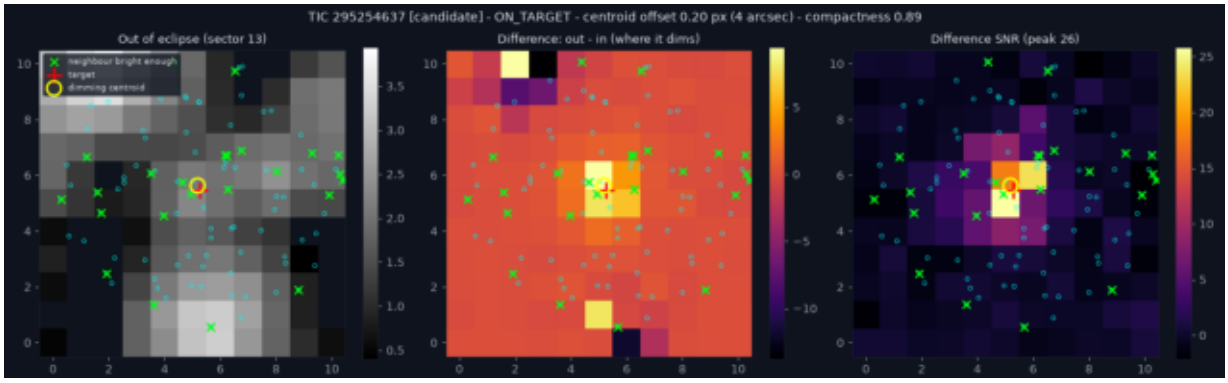
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 13    light-curve dip in this sector: 57.0 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   25.85   (gate >= 5.0)      Compactness   0.89   (gate >= 0.50)
  Dimming centroid offset     0.20 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 295254639
    0.40 px away, dTmag +0.13 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 15.37 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 5.329910 d, in any write-up.